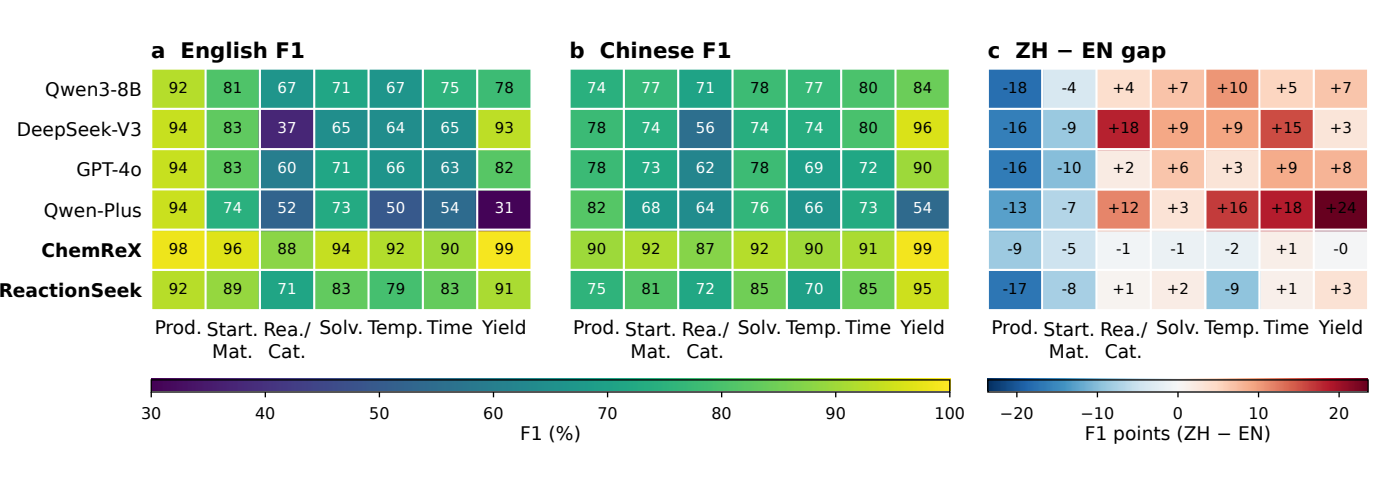

a English F1
b Chinese F1
c ZH − EN gap
Qwen3-8B
DeepSeek-V3
GPT-4o
Qwen-Plus
ChemReX
ReactionSeek
92 81 67 71 67 75 78
94 83 37 65 64 65 93
94 83 60 71 66 63 82
94 74 52 73 50 54 31
98 96 88 94 92 90 99
92 89 71 83 79 83 91
74 77 71 78 77 80 84
78 74 56 74 74 80 96
78 73 62 78 69 72 90
82 68 64 76 66 73 54
90 92 87 92 90 91 99
75 81 72 85 70 85 95
-18 -4 +4 +7 +10 +5 +7
-16 -9 +18 +9 +9 +15 +3
-16 -10 +2 +6 +3 +9 +8
-13 -7 +12 +3 +16 +18 +24
-9 -5 -1 -1 -2 +1 -0
-17 -8 +1 +2 -9 +1 +3
Prod.
Start. Mat.
Rea./ Cat.
Solv.
Temp.
Time
Yield
30 40 50 60 70 80 90 100
F1 (%)
−20 −10 0 10 20
F1 points (ZH − EN)